Hands-On-Activity No. 2.1

Data Types and Arithmetic Operations

Course Code: CPE 007	**Program: Computer Engineering**
Course Title: Programming logic and Design	**Date Performed: July 21, 2026**
Section: CPE11S2	**Date Submitted: July 23, 2026**
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1. Objective(s)

This activity aims to develop programming skills in structuring data types and arithmetic operations.

2. Intended Learning Outcomes

2.1 Find possible compilation errors and logic errors

2.2 Complete programs applying data types and arithmetic operations

3. Materials and Equipment

Internet connection
Personal Computer

4. Procedure

5. Output

1.) Take a look at the code below: it assigns two integer values, manipulates them and finally outputs the result and bigresult variables. The problem is that the manipulations have been described using natural language, so the code is completely useless now. Act as an intelligent (naturally!) compiler and translate the formula into a real "C" code notation. Test your code using the data provided.

From

```
#include <iostream>
using namespace std;
int main(void)
{
   int xValue=5;
   int yValue=9;
   int result;
   int bigResult;
 /*
        increment xValue by 3
        decrement yValue by xValue
        multiply xValue times yValue giving result
        increment result by result
        decrement result by 1
        assign result modulo result to yValue
        increment result by result added to xValue
        assign result times result times result to bigResult  increment result by xValue times yValue
        */
   cout<<"result: "<<result;
   cout<<"big result: "<< bigResult;
return 0;
}
```

To

```
#include <iostream>
        using namespace std;

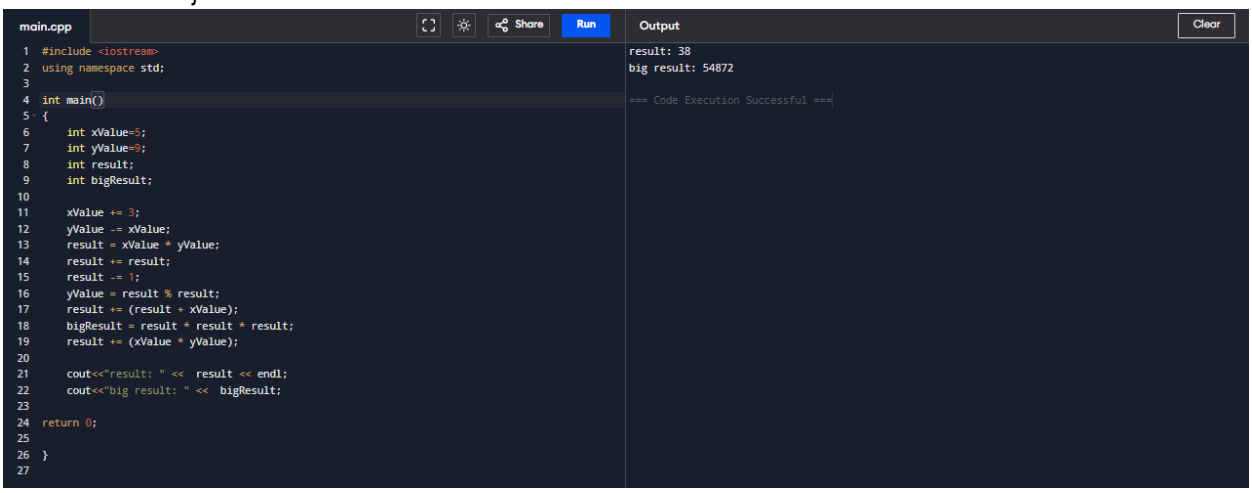
        int main()
        {
                int xValue=5;
                int yValue=9;
                int result;
                int bigResult;

                xValue += 3;
                yValue -= xValue;
                result = xValue * yValue;
                result += result;
                result -= 1;
                yValue = result % result;
                result += (result + xValue);
                bigResult = result * result * result;
                result += (xValue * yValue);

           cout<<"result: " <<  result << endl;
           cout<<"big result: " <<  bigResult;

        return 0;

        }
```



2. Complete the program below. Compute the accrued amount of money with a starting value of 100 and an annual interest rate of 1.5%. Compute and print the results for first three years. Your version of the program must print the same result as the expected output for every year. Compute each annual value on the basis of the previous year's value.

From

```
#include <iostream>
using namespace std;
int main(void)
{
   int xValue=5;
   int yValue=9;
   int result;
   int bigResult;
            /*
            increment xValue by 3
            decrement yValue by xValue
            multiply xValue times yValue giving result
            increment result by result
            decrement result by 1
            assign result modulo result to yValue
            increment result by result added to xValue
            assign result times result times result to bigResult  increment result by xValue times yValue
            */
   cout<<"result: "<<result;
   cout<<"big result: "<< bigResult;
return 0;
}
```

To

```
#include <iostream>
using namespace std;
int main()
{
                        float startValue = 100;
                        float interestRate = 0.015;
                        float firstYearValue;
                        float secondYearValue;
                        float thirdYearValue;
                        firstYearValue = startValue * (1 + interestRate);
                        secondYearValue = firstYearValue * (1 + interestRate);
                        thirdYearValue = secondYearValue * (1 + interestRate);

            cout << "After first year: " << firstYearValue << endl;
            cout << "After second year: " << secondYearValue << endl;
            cout << "After third year: " << thirdYearValue;
return 0;
}
```

main.cpp

Share Run

```cpp
#include <iostream>

using namespace std;

int main()

{
    float startValue = 100;
    float interestRate = 0.015;
    float firstYearValue;
    float secondYearValue;
    float thirdYearValue;
    firstYearValue = startValue * (1 + interestRate);
    secondYearValue = firstYearValue * (1 + interestRate);
    thirdYearValue = secondYearValue * (1 + interestRate);

cout << "After first year: " << firstYearValue << endl;
cout << "After second year: " << secondYearValue << endl;
cout << "After third year: " << thirdYearValue;

return 0;

}

```

Output

Clear

```
After first year: 101.5
After second year: 103.022
After third year: 104.568

=== Code Execution Successful ===
```

6. Conclusion

Learning how increment and decrement or rather the data types and arithmetic operations

7. Rubric